

Q 4

(i)

$$\text{Transmission Rate (R)} = 350 \text{ Mb/s}$$

$$\text{Object size (L)} = 250,000 \text{ bits}$$

$$\text{Request Rate (\lambda)} = 35 \text{ req/s}$$

$$\text{Avg internet-side round trip (T}_{\text{internet}}) = 2.5 \text{ sec}$$

$$(i) \text{ Total avg response time (T}_{\text{total}}) = ?$$

$$\begin{aligned} \text{Data rate (Arrival rate)} &= 250,000 \times 35 \\ &= 8.75 \times 10^6 \text{ bits/s} \\ &= 8.75 \text{ Mb/s} \end{aligned}$$

$$\text{Utilization } (\rho) = \frac{\text{Arrival rate}}{\text{Transmission rate}}$$

$$= \frac{8.75 \text{ Mb/s}}{350 \text{ Mb/s}}$$

$$= 0.025$$

$$= 2.5\%$$

$$\begin{aligned} \text{Access delay (T}_{\text{access}}) &= \frac{L/R}{1-\rho} \\ &= \frac{250,000 / (350 \times 10^6)}{1-0.025} \end{aligned}$$

$$= 0.73 \times 10^{-3}$$

$$\therefore T_{\text{total}} = T_{\text{access}} + T_{\text{internet}}$$

$$= (0.73 \times 10^{-3}) + 2.5$$

$$= 2.50073$$

$$\boxed{T_{\text{total}} \approx 2.5 \text{ Sec}} \text{ Ans.}$$

(ii)

$$\text{Miss rate} = 0.7$$

$$\text{Response time of cache} = 0.01$$

$$\Rightarrow \text{Hit rate} = 1 - 0.7 \\ = 0.3$$

$\Rightarrow$ Total response time (with cache):

$$T_{\text{total}(c)} = (\text{Miss rate} * T_{\text{total}}) + (\text{HitRate} * \text{Response cache time}) \\ = (0.7 * 2.50073) + (0.3 * 0.01)$$

$$T_{\text{total}(c)} = 1.753 \text{ sec} \text{ Ans.}$$

Q 5

Data:

$$\text{Buffer max capacity } (b_{\text{max}}) = 50\,000 \text{ packets}$$

$$\text{Packet size } (p_{\text{size}}) = 1000 \text{ bytes}$$

$$\text{Ongoing link} = 100 \text{ Mbps}$$

$$\text{Arrival link} = 150 \text{ Mbps}$$

$$t = 10 \text{ s}$$

1- Time for buffer overflow = ?

2- Packets dropped = ?

3- Advanced mechanisms (2) = ?

Sol:

$$\therefore 1 \text{ byte} = 8 \text{ bits}$$

$$b_{\text{max}} = 50\,000 \times (1000 \text{ bytes}) \times 8 \text{ bits} \\ = 400 \times 10^6 \text{ bits} \\ = 400 \text{ Mbits}$$

$$\text{Buffer overload} = (\text{Arrival link}) - (\text{Ongoing link})$$

$$= 150 - 100$$

$$V_{\text{overload}} = 50 \text{ Mbps}$$

$$\begin{aligned}\text{Overloaded data in } t \text{ secs} &= b_{\text{overload}} \times t \\ &= (50 \times 10) \text{ (Mbps} \times \text{s)} \\ &= 500 \text{ Mbits}\end{aligned}$$

$$\begin{aligned}\text{Time for buffer overflow} &= \frac{b_{\text{max}}}{V_{\text{overload}}} \\ &= \frac{400 \text{ Mbits}}{50 \text{ Mbps}} \\ 1. \quad \boxed{t_{\text{overflow}} = 8 \text{ s}} &\rightarrow \underline{\text{Ans.}}\end{aligned}$$

$$\begin{aligned}\text{Data dropped} &= (500 - 400) \text{ Mbits} \\ &= 100 \times 10^6 \text{ bits} \\ &= (100 \times 10^6) \div 8 \\ &= 12.5 \times 10^6 \text{ bytes}\end{aligned}$$

$$\text{Packets dropped} = (12.5 \times 10^6) \div 1000$$

$$2. \quad \boxed{= 12500 \text{ packets}} \underline{\text{Ans.}}$$

3- Advanced Mechanisms:

① Priority Queueing:

- Router maintains multiple queues.
- Different traffic goes into different queues.
- High priority packets get transmitted.
- When buffer fills, it drops low priority packets, first.

② Congestion Window Adjustment (At TCP Level):

- TCP sender maintains a congestion window (packets it can send)
- Starts small, then gradually increases.
- If packet loss detected, it immediately reduce window size.
- If no loss occurs, it gradually increases again.